# Problem Set 02

## Informed Search, Local Search, Games

**Course:** CSE 422: Artificial Intelligence **Total Marks:** 50

**Instructions:** Show all intermediate steps clearly. Justify every answer.

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### Problem 1 [7 Marks]

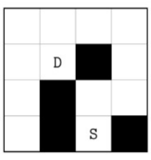
A rescue drone is deployed inside a damaged research facility represented as an $(n \times m)$ grid. The current location of the drone is marked by `D`, and the emergency supply station is marked by `S`. Some cells are blocked by collapsed structures and cannot be entered.

The drone may move only to the **top, bottom, left,** and **right** adjacent cells. Each move has unit cost.

Your task is to compute the shortest path from the drone's starting location to the supply station using the **A* search algorithm** with **Manhattan distance** as the heuristic function.

The top-left cell of the grid is $(1,1)$ and the goal cell is $(4,3)$. If $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are two cells, then the Manhattan distance between them is defined as

$$h(P_1, P_2) = |x_1 - x_2| + |y_1 - y_2|.$$




**Task.** Simulate the execution of A* search step by step.

At each iteration:

- show which node is removed from the fringe,
- list the newly generated nodes,
- show the full fringe with their corresponding

$$f(n) = g(n) + h(n)$$

values.

Continue the simulation until the goal node is removed from the fringe.

### Problem 2 [3 Marks]

Suppose you are given three heuristic functions $h_1$, $h_2$, and $h_3$. Assume that $h_1$ and $h_2$ are admissible, but $h_3$ is inadmissible.

You define the following new heuristic functions:

$$h_4(n) = 0$$
$$h_5(n) = 2h_2(n)$$
$$h_6(n) = \frac{h_1(n) + h_2(n)}{2}$$
$$h_7(n) = \max(h_1(n), h_2(n))$$





$$h_8(n) = \min(h_1(n), h_3(n))$$
$$h_9(n) = \max(h_2(n), h_3(n))$$

Answer the following questions. In each case, provide a brief but precise justification.

a. Which two heuristics are *possibly inadmissible*?

b. Between $h_6$ and $h_7$, which heuristic dominates the other?

c. In your opinion, which heuristic is the best choice for A* search among the above? Explain.

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### Problem 3 [10 Marks]

**True/False Questions**

Determine whether each of the following statements is **True** or **False**. Provide a brief but clear justification for each answer.

a. **[2.5 Marks]** In hill-climbing search, allowing sideways moves on a plateau increases the probability of reaching a global optimum.

b. **[2.5 Marks]** Simulated Annealing does not require knowledge of the entire search space to operate effectively.

c. **[2.5 Marks]** Hill-climbing algorithms cannot explore the search space exhaustively.

d. **[2.5 Marks]** You are programming a robot to navigate a simple maze. The robot uses a hill-climbing algorithm, meaning it always moves in the direction that appears to bring it closer to the exit based on local information. Using hill-climbing guarantees that the robot will find the shortest path to the exit.

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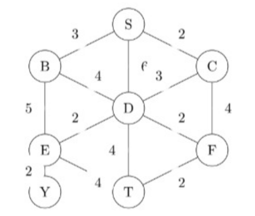
**Instruction:** Answers without justification will receive reduced marks.

### Problem 4 [8 Marks]

Consider the weighted graph below. Let **S** be the initial state and **T** be the goal state.

Two heuristic functions, $h_1(n)$ and $h_2(n)$, are given for most nodes. One node, **Y**, has missing heuristic values.




| **State, $n$** | $h_1(n)$ | $h_2(n)$ |
|---|---|---|
| S | 6 | 5 |
| B | 5 | 4 |
| C | 4 | 3 |
| D | 3 | 2 |
| E | 4 | 3 |
| F | 2 | 2 |
| Y | ?? | ?? |
| T | 0 | 0 |

a. **[2 Marks]** Examine node **Y** and assign reasonable heuristic value(s) to it. Justify your choice.





b. **[4 Marks]** Assume both $h_1$ and $h_2$ are admissible. Apply **custom A* search** on the graph using the *better* heuristic. In the search, if a node already in the fringe is found again with a lower $f$-cost, replace the older copy in the fringe. Show the steps clearly.

c. **[1 Mark]** Suppose **C** is the initial state and **E** is the goal state. Determine whether A* search using either $h_1$ or $h_2$ will return the optimal path from C to E.

d. **[1 Mark]** Now suppose the edge cost from **B** to **E** becomes 3 and the edge cost from **B** to **D** becomes 2. Determine whether the heuristic value $h_1(B)$ remains **consistent**. Show your calculations.

### Problem 5 [10 Marks]

**Game Tree Search with Alpha-Beta Pruning**

An AI agent (**Player O**) is playing a strategic $3 \times 3$ grid-based game against a human opponent (**Player X**). The current configuration of the board is shown below.

| | | |
|---|---|---|
| O | O | X |
| | X | |
| O | X | |

The agent must decide its next move using the **Minimax algorithm with Alpha-Beta pruning**. The utility function is defined as follows:

$$U = \begin{cases} (100 - d) \times (+1), & \text{if the agent wins} \\ (100 - d) \times (-1), & \text{if the agent loses} \\ 0, & \text{if the game results in a draw} \end{cases}$$

where $d$ is the depth of the terminal node.

*Assume the root node is at depth 1.*

a. **[7 Marks]** Construct the game tree from the current state and determine the move that the agent will choose.

- Clearly show the expansion of the tree.
- Apply **alpha-beta pruning** where possible.
- Annotate each node with its computed utility value.
- Highlight the optimal move selected by the agent.

b. **[3 Marks]** Consider the statement:

"Any state of this game can be used to demonstrate that the environment is zero-sum."

Evaluate the correctness of this statement.

- Use **two different game states** from your constructed tree.
- Show mathematically that the sum of utilities for both players is zero.
- Provide a concise justification.

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**Submission guideline:** Clearly structured trees and proper pruning justification are required for full credit.